

2. IDP MFA (Multi-Factor Authentication) Management – SQL Reference



Purpose:

This page provides SQL Server scripts to view, troubleshoot, and manage MFA (PIN-based) authentication records stored in the Token . AuthSession table for IDP environments. These queries are intended for QA/Stage troubleshooting only — do not run in Production.

Environments

Environment	Server	Database	Authentication	Username
Stage1	GWQMSSQL02.rsd.crumprsd.com	AGSAuthStage1	Windows Authentication	RSD\<your username>
QC	GWQMSSQL01.rsd.crumprsd.com	AGSAuthQC4, AGSAuthQC1	Windows Authentication	RSD\<your username>

Table: Token.AuthSession

Purpose: Stores temporary MFA PINs and session data for authentication.

Key Columns:

- AuthSessionId – Unique session identifier
- SubjectId – User identifier
- Pin – Contains the MFA PIN and timestamp (e.g., 684830, 10/9/2025 9:16:37 AM)
- ClientId – Client reference key
- CreatedOn / ModifiedOn – Session timestamps

Behavior Notes:

- When MFA PIN is generated Pin column stores "PIN, Timestamp".
- When PIN is used or expires Pin becomes '' (empty) or NULL.

Common SQL Queries

1 Get Latest Active PIN for a User

Returns the latest unused MFA PIN for a specific user.

```
SELECT TOP 1 AuthSessionId, SubjectId, LEFT(Pin, CHARINDEX(',', Pin + ',') - 1) AS ActualPin, CreatedOn, ClientId
FROM Token.AuthSession
WHERE SubjectId = 'Test_Dinesh001' AND Pin IS NOT NULL AND LTRIM(RTRIM(Pin)) <> ''
ORDER BY CreatedOn DESC;
```

2 Get All Active PINs Generated Today

```
SELECT AuthSessionId, SubjectId, LEFT(Pin, CHARINDEX(',', Pin + ',') - 1) AS ActualPin, CreatedOn
FROM Token.AuthSession
WHERE Pin IS NOT NULL AND LTRIM(RTRIM(Pin)) <> ''
AND CAST(CreatedOn AS DATE) = CAST(GETDATE() AS DATE)
ORDER BY CreatedOn DESC;
```

3 View All PINs and Status

```
SELECT AuthSessionId, SubjectId,
CASE WHEN LTRIM(RTRIM(Pin)) = '' THEN 'USED/EXPIRED'
      WHEN Pin IS NULL THEN 'NULL'
      ELSE 'ACTIVE' END AS PinStatus,
Pin, CreatedOn
FROM Token.AuthSession
ORDER BY CreatedOn DESC;
```

4 Find Cleared (Used/Expired) PINs

```
SELECT AuthSessionId, SubjectId, Pin, CreatedOn
FROM Token.AuthSession
WHERE Pin = '' OR Pin IS NULL
ORDER BY CreatedOn DESC;
```

5 Find Active PINs by Client ID

```
SELECT AuthSessionId, SubjectId, LEFT(Pin, CHARINDEX(',', Pin + ',') - 1) AS ActualPin, ClientId, CreatedOn
FROM Token.AuthSession
WHERE ClientId = '7DF0C0292F9445CB99AB0103618A9901'
    AND Pin IS NOT NULL AND LTRIM(RTRIM(Pin)) <> ''
ORDER BY CreatedOn DESC;
```

6 Count Active PINs per User

```
SELECT SubjectId, COUNT(*) AS ActivePinCount
FROM Token.AuthSession
WHERE Pin IS NOT NULL AND LTRIM(RTRIM(Pin)) <> ''
GROUP BY SubjectId
ORDER BY ActivePinCount DESC;
```

7 Extract Last PIN Generated for Each User

```
SELECT t.SubjectId, LEFT(t.Pin, CHARINDEX(',', t.Pin + ',') - 1) AS LastPin, t.CreatedOn
FROM Token.AuthSession t
INNER JOIN (
    SELECT SubjectId, MAX(CreatedOn) AS LatestCreated
    FROM Token.AuthSession
    GROUP BY SubjectId
) latest ON latest.SubjectId = t.SubjectId AND latest.LatestCreated = t.CreatedOn;
```

8 Find PINs Generated in Last 24 Hours

```
SELECT AuthSessionId, SubjectId, LEFT(Pin, CHARINDEX(',', Pin + ',') - 1) AS ActualPin, CreatedOn
FROM Token.AuthSession
WHERE Pin IS NOT NULL AND LTRIM(RTRIM(Pin)) <> ''
    AND CreatedOn >= DATEADD(HOUR, -24, GETDATE())
ORDER BY CreatedOn DESC;
```

9 Audit PIN Creation & Modification

```
SELECT AuthSessionId, SubjectId, Pin, CreatedOn, ModifiedOn, ModifiedBy
FROM Token.AuthSession
WHERE Pin IS NOT NULL AND LTRIM(RTRIM(Pin)) <> ''
ORDER BY CreatedOn DESC;
```

Optional Cleanup – Old Empty Sessions

```
-- Remove test MFA sessions older than 30 days (use with caution)
DELETE FROM Token.AuthSession
WHERE (Pin = '' OR Pin IS NULL) AND CreatedOn < DATEADD(DAY, -30, GETDATE());
```

Usage Guidelines

1. Connect to your desired environment (Stage1 or QC) using **Windows Authentication**.
2. Open a **new query window** in SSMS and run individual queries as needed.
3. Always review filters like SubjectId or ClientId before executing.
4. Avoid DELETE statements unless cleaning up expired test data.
5. For debugging MFA issues:
 - Check latest PIN via Query 1
 - Verify if PIN is cleared post-login (Query 4)
 - Correlate timestamps with user login attempts